Station-485

SKU:K123





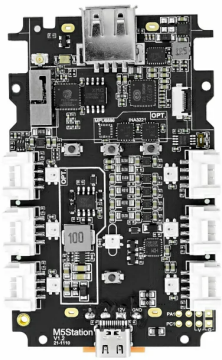














Description

Station-485 is a versatile **industrial-grade** programmable embedded controller. It uses the Espressif **ESP32** as the main control chip, integrates a **Wi-Fi** solution, and is equipped with a dual-core low-power **Xtensa® 32-bit LX6** microprocessor with a clock speed of up to **240MHz**. It comes with **16MB Flash** and integrates a **240 x 135 1.14-inch full-color HD IPS display panel**, **physical button panel**, **rich peripherals**, and **two sets of six expansion interfaces**. It also features **low-power sleep/timed wake-up** functionality.

It supports multiple power supply methods, including **USB Type-C, PWR485, and an internal rechargeable battery (battery socket reserved on the board)**. The board integrates a high-power-density fully integrated boost DC/DC converter SCT12A0DHKR, ensuring **stability** in complex application scenarios.

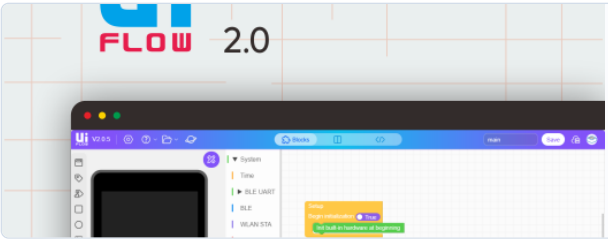
This controller is suitable for various application scenarios such as industrial field control, smart buildings, multi-channel data acquisition nodes, and developer prototyping.

Tutorial



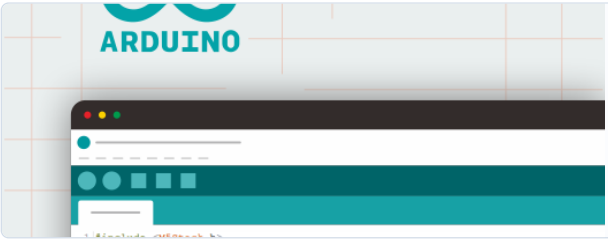
UIFlow

This tutorial will introduce how to control the Station device via the UIFlow graphical programming platform



UiFlow2

This tutorial will introduce how to control the Station device via the UiFlow2 graphical programming platform



Arduino IDE

This tutorial will introduce how to program and control the Station device using the Arduino IDE.

Features

- **Interactive Design**:
  - 1.14-inch IPS display panel
  - 3 programmable physical buttons
  - 1 power button
  - 7 programmable RGB LEDs
- **Power Design**:
  - Input:
    - Integrated 9~24V to 5V DC/DC **SY8303** step-down circuit
    - **AXP192** power management chip
  - Output:
    - Each interface (5 Grove, 1 USB-A) uses an electronic switch **SGM2553D** for independent on/off control
    - 6 Grove interfaces use **INA3221** for voltage/current monitoring, and USB-A uses **INA199** for current monitoring
    - Integrated high-power-density fully integrated boost DC/DC converter **SCT12A0DHKR**
  - Low Power Consumption:
    - Integrated RTC **BM8563** for low-power sleep/timed wake-up
- **Port Design**:
  - 6 Grove expansion interfaces
    - Port A1/A2 share power and signal pins
    - Port B1/B2/C1/C2 have independent power and signal pins
  - **PWR485** interface (4-wire HT3.96)
    - RS485

        - 9~24V power input
    - USB Type-A is for power output only, with no signal pins
- **Structural Design**:
    - **DIN rail**
    - **Magnetic**
    - **Wall-mounted**
    - **Screws**
    - **Cable ties**
- Development Platform
    - UiFlow1
    - UiFlow2
    - Arduino IDE
    - ESP-IDF
    - PlatformIO

Includes

- 1 x Station-485
- 1 x USB Type-C cable (100cm)
- 1 x Hex key
- 1 x I/O Sticker

Applications

- IoT controller
- Multi-channel data acquisition
- IoT product prototyping
- DIY projects

Specifications

Specification	Parameters
SoC	ESP32-D0WDQ6-V3 @ Xtensa® 32-bit LX6 dual-core processor, clock frequency 240MHz
DMIPS	600
SRAM	520KB
Flash	16MB
Wi-Fi	2.4 GHz Wi-Fi
USB Power	5V@1A
RS485 Power	9 ~ 24V@1A
Host Ports	USB Type-C x 1, HY2.0-4P (I2C+I/O+UART) x 6, Full-Size USB Type-A (OUTPUT), PWR485
LED	SK6812 x 7
Buttons	Power button, physical buttons x 3
IPS LCD Screen	1.14"@240 x 135 ST7789V2
RTC	BM8563
PMU	AXP192
Voltage/Current Monitor	INA3221 + INA199
USB Chip	CH9102F
TTL-RS485	SP3485
Step-Down Chip	SY8303
DC/DC Boost	SCT12A0DHKR
Power Distribution Switch	SGM2553D
Antenna	2.4G 3D antenna
Operating Temperature	0 ~ 60°C
Base Screw Specification	M2 x 8 hex socket screws
Internal PCB Reserved Interfaces	Battery interface (1.25mm-4P), USB line interface (1.25mm-4P)
Housing Material	ABS+PC
Product Size	88.0 x 65.0 x 35.0mm
Product Weight	63.6g
Package Size	104.0 x 73.0 x 54.5mm
Gross Weight	105.3g

Learn

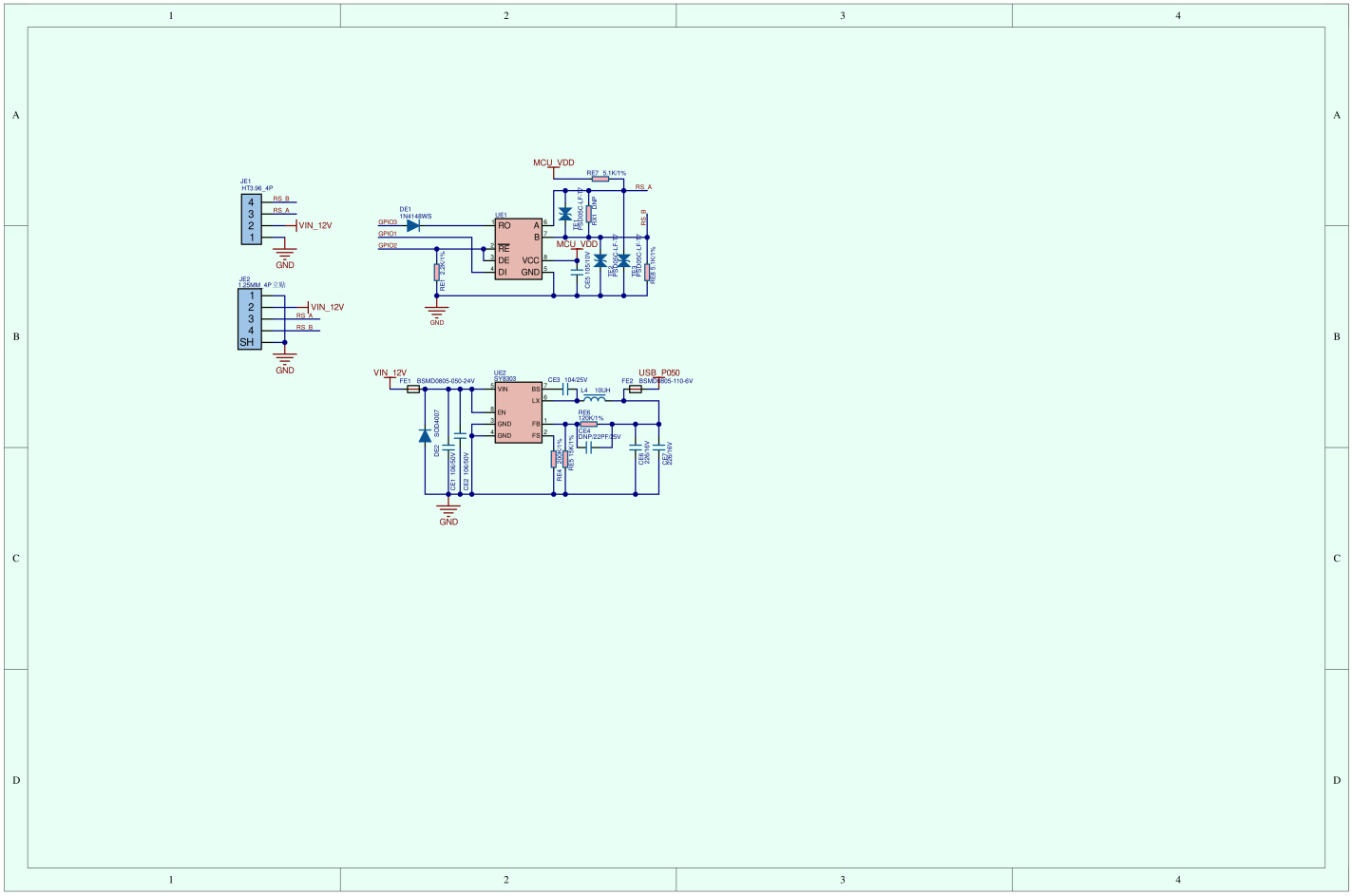
Power On/Off Operation:

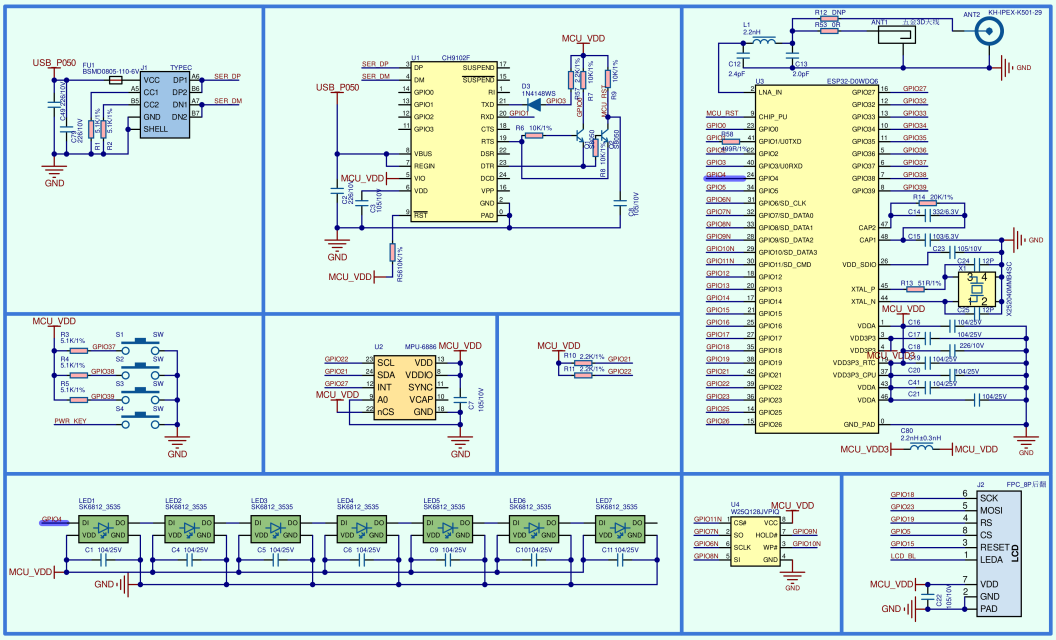
Power On: Click the central power button

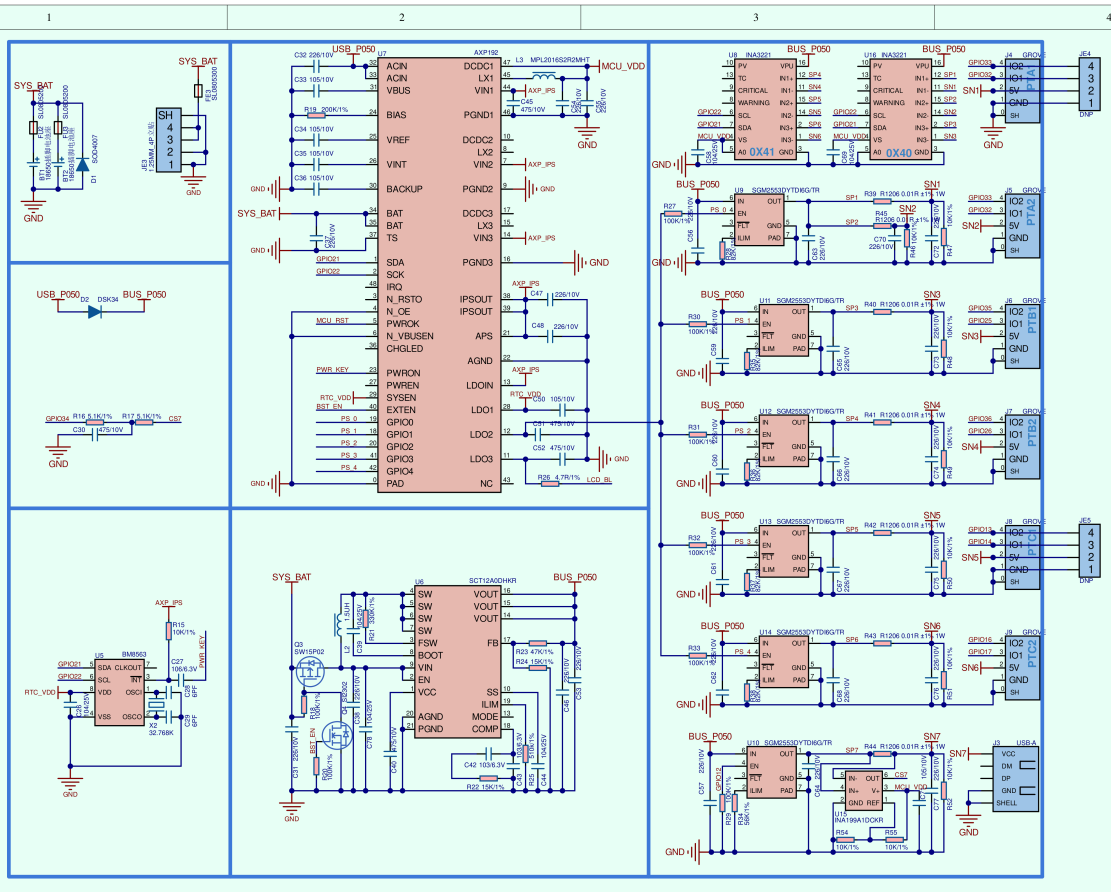
Power Off: Long press the central power button for 4 seconds

Schematics

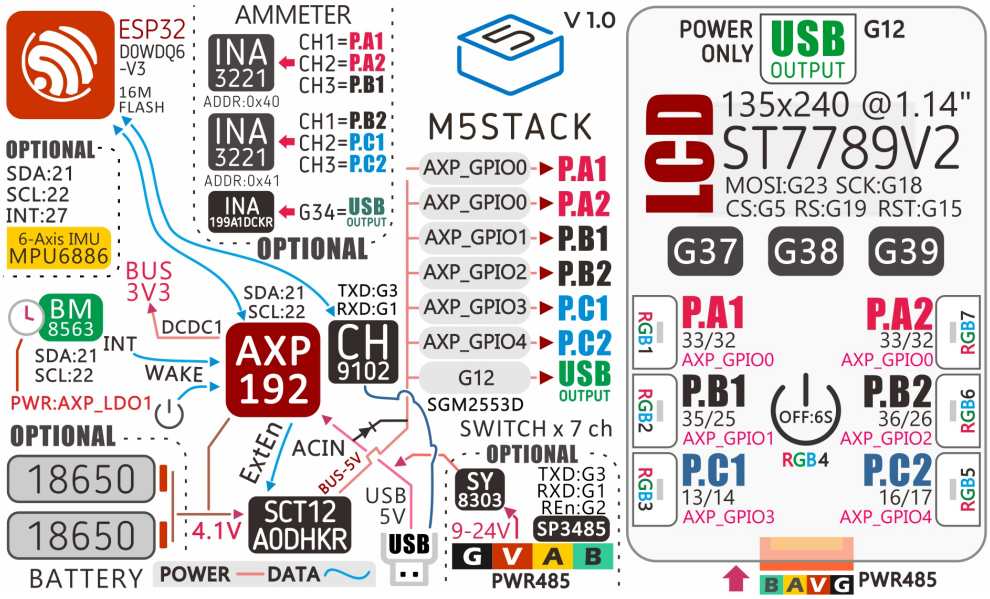
- Station-485 Schematics PDF







PinMap



Button

ESP32-D0WDQ6-V3	G37	G38	G39
Buttons	BUTTON A	BUTTON B	BUTTON C

Color TFT Screen

Driver Chip: ST7789v2 Resolution: 135 x 240 @1.14"

ESP32-D0WDQ6-V3	G5	G15	G18	G19	G23
AXP192 Chip					
LCD	CS(Chip Select)	RESET	SCK(SCLK)	RS(Date/Command)	MOSI

AXP192 Chip	AXP_LDO3
LCD	LCD_BL

RTC

ESP32-D0WDQ6-V3	G21	G22
BM8563	SDA	SCL
AXP192		

AXP192	AXP_PWR
BM8563	INT

Current/Voltage Monitor

ESP32-D0WDQ6-V3	G21	G22	Control Channel
INA3221 (0x40)	SDA	SCL	AXP_GPIO0, AXP_GPIO1
INA3221 (0x41)	SDA	SCL	AXP_GPIO2, AXP_GPIO3, AXP_GPIO4

Internal I2C Connection

ESP32-D0WDQ6-V3	G21	G22
AXP192(0x34)	SDA	SCL
BM8563(0x51)	SDA	SCL
INA3221(0x40、0x41)	SDA	SCL

PWR485

Chip	G3	G1	G2	(DC-DC 9~24V to 5V)	GND
SP3485	TXD	RXD	Send-High Active/Receive-Low Active		GND
SY8303				VIN_12V	

Power Management Chip (AXP192)

RTC	LCD BackLight	ESP32-3.3V SK6812, INA3221, CH902F
LDO1	LDO3	DC-DC1

USB to Serial

ESP32-D0WDQ6-V3	G3	G1
CH9102F	TXD	RXD

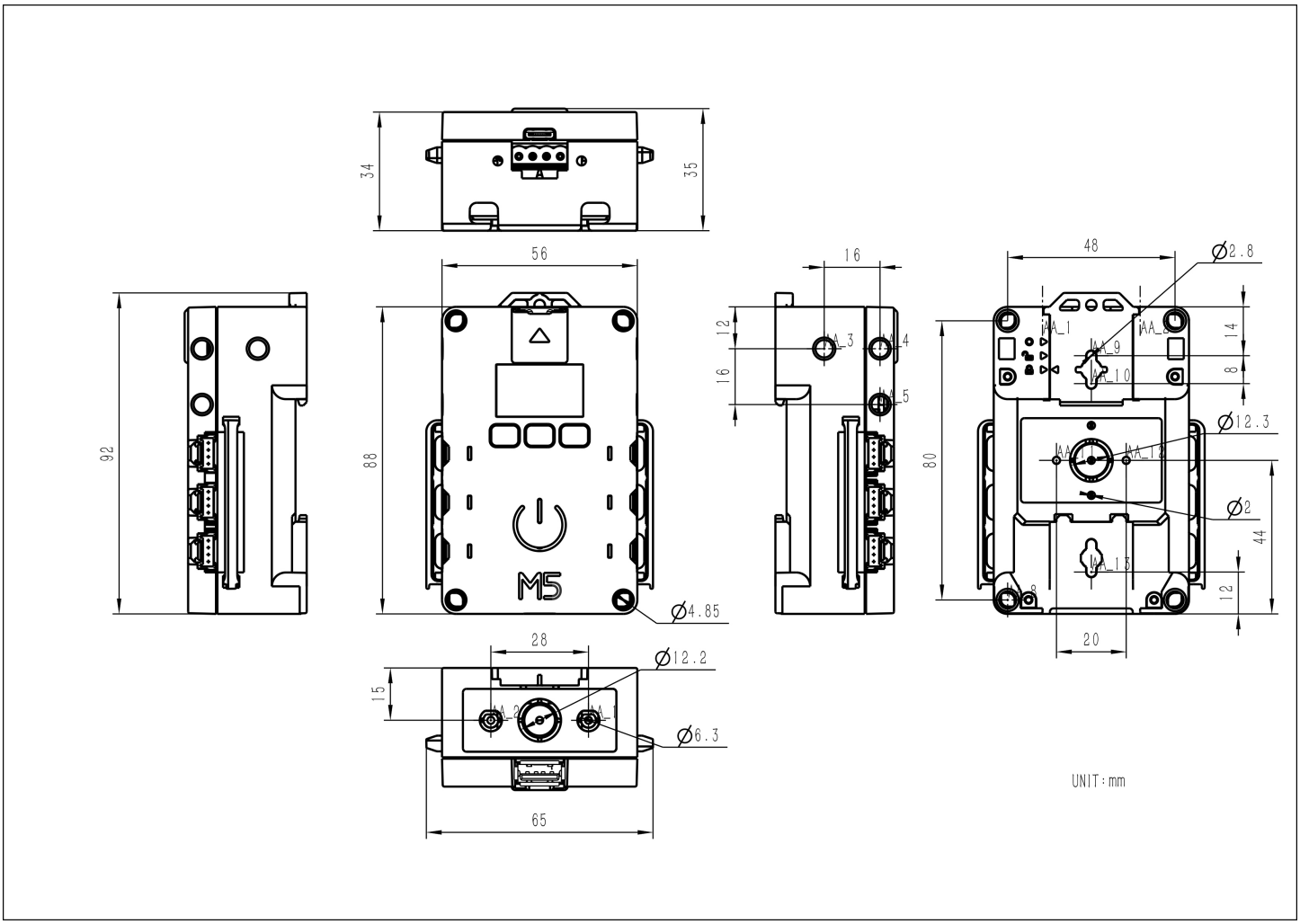
HY2.0-4P

HY2.0-4P	Black	Red	Yellow	White
PORT.A1	GND	5V	G32	G33
PORT.A2	GND	5V	G32	G33
PORT.B1	GND	5V	G25	G35
PORT.B2	GND	5V	G26	G36
PORT.C1	GND	5V	G14	G13
PORT.C2	GND	5V	G17	G16

Power Consumption Test

Operating Current (LCD, LED, Wi-Fi On)	Standby Current (LCD, LED On)
161mA	105mA

Model Size



Structure

- Station-485 Structure Files

Datasheets

- ESP32
- ST7789v2
- SY8303AIC
- SP3485
- SCT12A0DHKR
- INA3221_CN
- INA199_CN
- SGM2553
- BM8563
- AXP192 datasheet
- AXP192 register

Softwares

Arduino

- Station-485 Arduino Quick Start
- Station-485 Arduino Driver Library

UiFlow1

- Station UiFlow1 Quick Start

UiFlow2

- Station UiFlow2 Quick Start

USB Driver

> Click the link below to download the driver for your operating system. After extracting the compressed file, select the installation package corresponding to your operating system's bit version. (**CH9102_VCP_SER_MacOS v1.7** may report an error during installation, but it is actually installed successfully. Ignore the error.) If you encounter issues such as timeout or "Failed to write to target RAM" when downloading the program, try reinstalling the device driver.

Driver Name	Applicable Driver Chip	Download Link
CH9102_VCP_SER_Windows	CH9102	Download
CH9102_VCP_SER_MacOS v1.7	CH9102	Download

Easyloader

Easyloader	Download Link	Remarks
Station-485 Test Easyloader	Download	/

Product Comparison

Station-485	Station-BAT
Equipped with PWR485 (RS485 + Power Input)	Onboard MPU6886, can be equipped with two 18650 batteries (parallel)

To compare information on the Station series products, you can visit the Product Selection Table, check the target products, and get the comparison results. The selection table covers key information such as core parameters and functional features, and supports comparison of multiple products simultaneously.